Kessler Mobility

Technical Specification

RFQ KM-2026-0231 - Inverter Gen3 programme

Issued to supplier: Acme Components

Issue date: 2026-09-21

Response due: 2026-11-06

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This document is a fictional sample created for software testing. All names, figures and references are invented. Standards are cited by name and number only.

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Each numbered clause states one requirement. The Supplier answers every clause in the questionnaire and in the response letter using one of five statuses.

Section 1 - Scope and definitions

This section defines the scope of the request and the terms used in this document.

1.1 This specification applies to all parts for the Kessler Mobility Inverter Gen3 programme that Acme Components is invited to quote under RFQ KM-2026-0231, covering the six part families listed in section 2.

1.2 In this document Buyer means Kessler Mobility, Supplier means Acme Components, and Part means any item supplied to a Buyer drawing or purchase order issued under this RFQ.

1.3 The Buyer documents prevail over every Supplier document, and Supplier quotation conditions, general terms and reservations have no effect unless the Buyer accepts them in writing.

1.4 The Supplier quotation shall remain valid for 180 calendar days from the response due date.

1.5 Start of production is planned for March 2028; the Supplier shall confirm tooling and capacity for this date.

Section 2 - Technical requirements

Numeric values in this section are minimum performance levels unless stated otherwise. Sub-sections follow the six part families in scope.

Sub-section HSG - Aluminium die-cast housings

2.1 Aluminium die-cast inverter housings shall operate continuously from -40 C to +150 C.

2.2 Housings for the silicon carbide power stage shall operate continuously at a coolant-side temperature of +175 C.

2.3 Housing castings shall achieve a tensile strength of at least 280 MPa.

2.4 Housings shall withstand 720 h of neutral salt spray with no corrosion penetrating the coating.

2.5 Sealed housings shall meet ingress protection IP6K9K in the fully assembled condition.

2.6 The sealing face of each housing shall be flat within 0.15 mm and within 0.05 mm over any 25 mm length.

2.7 Coolant jackets shall pass a leak test at the pressure and medium defined in the Buyer test sheet, which the Buyer will issue after nomination.

Sub-section FST - Forged steel fasteners class 10.9

2.8 Class 10.9 forged steel fasteners shall have a tensile strength of at least 1040 MPa.

2.9 Fasteners shall carry a zinc-nickel coating applied to the Buyer coating specification.

2.10 Coated fasteners shall withstand 1000 h of salt spray to red rust.

2.11 Fasteners shall retain their rated properties from -40 C to +200 C.

2.12 The Supplier shall provide hydrogen embrittlement baking records for every coating lot with a retained test sample.

Sub-section SNS - PA66-GF30 sensor housings

2.13 PA66-GF30 sensor housings shall operate from -40 C to +125 C.

2.14 Sensor housings shall meet ingress protection IP69K.

2.15 Metallic inserts and coated surfaces of sensor housings shall withstand 240 h of salt spray.

2.16 Sensor housings shall be moulded from glass-fibre reinforced polyamide 66 with 30 percent glass content.

2.17 Sensor housings shall pass the vibration profile in the Buyer test specification, which is still in draft.

Sub-section CON - Terminal and connector assemblies

2.18 Terminal and connector assemblies shall operate from -40 C to +125 C.

2.19 Connector assemblies shall meet ingress protection IP67 when mated.

2.20 Connector assemblies shall carry a rated current of 30 A, and a busbar variant shall be rated at 50 A.

2.21 Connector assemblies shall withstand 10000 cycles of mating and unmating.

2.22 Connector assemblies of the high-voltage interface shall meet ingress protection IP69K when mated.

Sub-section BRK - Ductile iron brake caliper brackets

2.23 Ductile iron brake caliper brackets shall achieve a tensile strength of at least 500 MPa.

2.24 Brackets shall withstand 480 h of salt spray.

2.25 Brackets mounted near the brake disc shall operate at temperatures up to +220 C.

2.26 The Supplier shall provide fatigue life data for brackets to the Buyer load spectrum.

2.27 The bracket coating system shall be agreed with the Buyer paint shop before the first sample submission.

Sub-section HRN - Wiring harness sub-assemblies

2.28 Wiring harness sub-assemblies shall operate from -40 C to +105 C.

2.29 Harness connectors shall meet ingress protection IP67 at the connector.

2.30 Every harness sub-assembly shall pass a 100 percent end-of-line electrical test before shipment.

2.31 The end-of-line test shall include a functional check at both temperature limits of the harness.

2.32 Harness sub-assemblies for the e-axle zone shall operate continuously at +150 C.

Section 3 - Materials and environmental

This section covers material declarations and environmental obligations.

3.1 The Supplier shall submit a material data sheet through the IMDS system for every Part before first sample submission.

3.2 All Parts shall comply with the REACH and RoHS regulations and the Supplier shall provide a written declaration on request.

3.3 The Supplier shall maintain an environmental management system certified to ISO 14001 at every producing site.

3.4 Aluminium castings shall contain at least 50 percent recycled content; the Buyer will define whether pre-consumer scrap counts.

3.5 The Supplier shall provide a product carbon footprint declaration for each part family once a year.

3.6 Packaging materials shall be recyclable and free of single-use plastics where technically feasible.

Section 4 - Quality

This section covers quality systems, certifications and quality performance.

4.1 The Supplier shall hold a valid IATF 16949 certificate covering every site that produces Parts under this RFQ.

4.2 The Supplier shall hold a valid ISO 9001 certificate for all producing sites.

4.3 The Supplier shall hold a valid ISO 45001 certificate for occupational health and safety.

4.4 The Supplier shall hold a valid TISAX AL3 label for information security before the nomination decision.

4.5 The Supplier shall provide an ISO 26262 process certificate for the development of safety-related Parts.

4.6 A third production plant shall be certified to IATF 16949 before start of production as a back-up source.

4.7 Production part approval shall be submitted at PPAP level 5 for all Parts, including on-site review at the Supplier.

4.8 PPAP level 3 documents shall be submitted within ten days of the first sample submission.

4.9 Delivered quality shall be below 25 PPM, measured monthly per part number.

4.10 The Supplier shall deliver a complete 8D report within 5 working days of a quality complaint.

4.11 The Supplier shall submit a control plan and a process FMEA for each Part with the PPAP package.

Section 5 - Logistics

This section covers lead times, delivery terms and data exchange.

5.1 The tooling lead time from tooling order to first sample shall not exceed 12 weeks.

5.2 The series lead time from release to delivery shall not exceed 6 weeks.

5.3 The Supplier shall deliver under Incoterms 2020 FCA at the named Supplier site.

5.4 The Supplier shall deliver all series Parts under Incoterms 2020 DDP, including import duties at the Buyer plants.

5.5 The Supplier shall also quote delivery under Incoterms 2020 DAP at the Buyer assembly plant.

5.6 The Supplier shall exchange delivery schedules and despatch advices with the Buyer through EDIFACT EDI messages.

5.7 In addition to EDIFACT messages, the Supplier shall confirm each order line through the Buyer web portal API.

5.8 Parts shall be shipped in returnable packaging defined in the Buyer packaging standard, which will be issued after nomination.

5.9 The Supplier shall absorb volume changes of plus or minus 20 percent within the series lead time.

Section 6 - Commercial and legal

This section covers commercial and legal conditions. The commercial terms document restates them.

6.1 Liquidated damages for late delivery shall be charged at the actual loss of the Buyer, without a cap.

6.2 The Supplier shall grant an annual price reduction of 4 percent for 5 years from start of production.

6.3 Payment shall be made 90 days net from the date of a correct invoice.

6.4 The Supplier warrants Parts against defects for 60 months with no mileage limit.

6.5 Supplier liability includes unlimited liability for indirect and consequential losses, including vehicle recall costs.

6.6 Tooling paid for by the Buyer remains the property of the Buyer and shall be marked as such.

6.7 The Supplier grants the Buyer a perpetual, free licence to the Supplier background intellectual property used in the Parts.

6.8 The Supplier shall accept the Buyer Supplier Code of Conduct in its current version.

6.9 The Buyer may audit the Supplier and its sub-suppliers on 10 working days notice.

6.10 The Supplier shall disclose its cost breakdown on request; the Buyer will define the level of detail in a template.

Annex A - Document register

The documents below are referenced by this specification and are available on the Buyer supplier portal after the supplier registers.

Document	Title	Revision
DOC-3108	Part drawings package	R6
DOC-3474	Quality manual requirements	R5
DOC-2609	Packaging guideline	R4
DOC-3782	Logistics manual	R4
DOC-5865	Supplier code of conduct	R2
DOC-8472	Test plan index	R5

Annex B - Submission instructions

Step 1: Submit the response through the Buyer portal before the due date shown on the cover page.

Step 2: Provide one response letter, the completed questionnaire workbook and the commercial terms mark-up.

Step 3: Answer every clause using exactly one status: Comply, Partial, Deviate, Exception or Clarify.

Step 4: Where a status is not Comply, state the reason and, where possible, the alternative offered.

Step 5: Questions about this RFQ must be sent in writing to the Buyer contact on the cover page.

Step 6: Answers to questions are shared with all invited suppliers without naming the asker.

Step 7: Reference your submission with the supplier number SUP-50961.

Annex C - Evaluation criteria

Offers are scored on the weighted criteria below after the technical clarification round.

Criterion	Weight
Technical compliance	35 percent
Quality systems	20 percent
Commercial terms	30 percent
Logistics and delivery	15 percent

Annex D - Revision history

Revision	Date	Change
R0	2026-09-21	First issue

Annex E - Glossary

8D - Eight-discipline problem solving report format.

EDIFACT - Electronic data interchange message standard family.

IMDS - Material data system used to declare vehicle materials.

PPAP - Production part approval process, submitted at levels 1 to 5.

PPM - Defective parts per million delivered.

TISAX - Information security assessment and exchange mechanism.